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# Modern Data Analysis — Task 2

**Student:** Odumosu Oluwasomidotun Adebowale
**Student No.:** 9
**Assigned Tasks:** 4, 8, 11, 17, 19, 27, 28, 33, 38, 42
**Course:** Modern Data Analysis
**Program:** Computational Modeling in Technology and Finance
**Lecturer:** Professor Prozorov O.A.

## 1. NumPy Setup

i. I used NumPy methods only to perform the calculations. I did not use standard Python loops, generators, or math functions. I also set a fixed random seed so that the results can be reproduced when the notebook is run again.

### ii. Task Information

This notebook contains my answers to the task numbers corresponding to the exercises assigned to Student No. 9 in the GroupList.

[56] ✓ 0s

```
import numpy as np

np.set_printoptions(precision=4, suppress=True)
```

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[56]
```
import numpy as np

np.set_printoptions(precision=4, suppress=True)
np.random.seed(42)

print("NumPy version:", np.__version__)
print("Random seed: 42")
```

```
NumPy version: 2.1.3
Random seed: 42
```

## Task 2.29 - Creating 1D and 2D Arrays

### 4. Create a vector with `zeros`

Method: I used the integer data type so that the vector is displayed as integer zeros.

[57]
```
vector_zeros = np.zeros(5, dtype=int)

print("Vector:")
print(vector_zeros)
```

```
Vector:
[0 0 0 0 0]
```

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## Task 2.30 - The random Module

### 8. Create a 2D array with `normal`

Method: I used [X \sim N(0,1),] where the mean is 0 and the standard deviation is 1. The array has 3 rows and 4 columns.

```
normal_array = np.random.normal(
    loc=0.0,
    scale=1.0,
    size=(3, 4)
)

print("2D normal-distribution array:")
print(normal_array)
```

```
2D normal-distribution array:
[[ 0.4967 -0.1383  0.6477  1.523 ]
 [-0.2342 -0.2341  1.5792  0.7674]
 [-0.4695  0.5426 -0.4634 -0.4657]]
```

## Task 2.31 - Array Attributes and Operations

### 11. Perform arithmetic operations on arrays

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Method: For arrays with the same shape, NumPy carries out addition, subtraction, multiplication, and division element by element.

[59] 0s

```
A11 = np.array([
    [2, 4, 6],
    [1, 3, 5]
])

B11 = np.array([
    [1, 2, 3],
    [4, 5, 6]
])

print("A + B:")
print(A11 + B11)

print("\nA - B:")
print(A11 - B11)

print("\nA * B:")
print(A11 * B11)

print("\nA / B:")
print(A11 / B11)
```

```
A + B:
[[ 3  6  9]
 [ 5  8 11]]
```

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```
A + B:
[[ 3  6  9]
 [ 5  8 11]]

A - B:
[[ 1  2  3]
 [-3 -2 -1]]

A * B:
[[ 2  8 18]
 [ 4 15 30]]

A / B:
[[2.     2.     2.    ]
 [0.25   0.625  0.8333]]
```

## Task 2.32 - Distance, Norm, Dot Product

17. Compute $\|a\|_\infty$ norm.

I computed the absolute value of every element in the vector $a$ and selected the largest value, the infinity norm is:

[ |a|_{\infty}=\max_i |a_i|. ]

```
a17 = np.array([3, -4, 5, -2])
```

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[60]

```
a17 = np.array([3, -4, 5, -2])

infinity_norm = np.linalg.norm(a17, ord=np.inf)

print("Vector:")
print(a17)
print("Infinity norm:", infinity_norm)
```

```
Vector:
[ 3 -4  5 -2]
Infinity norm: 5.0
```

## Task 2.33 - Matrix Operations

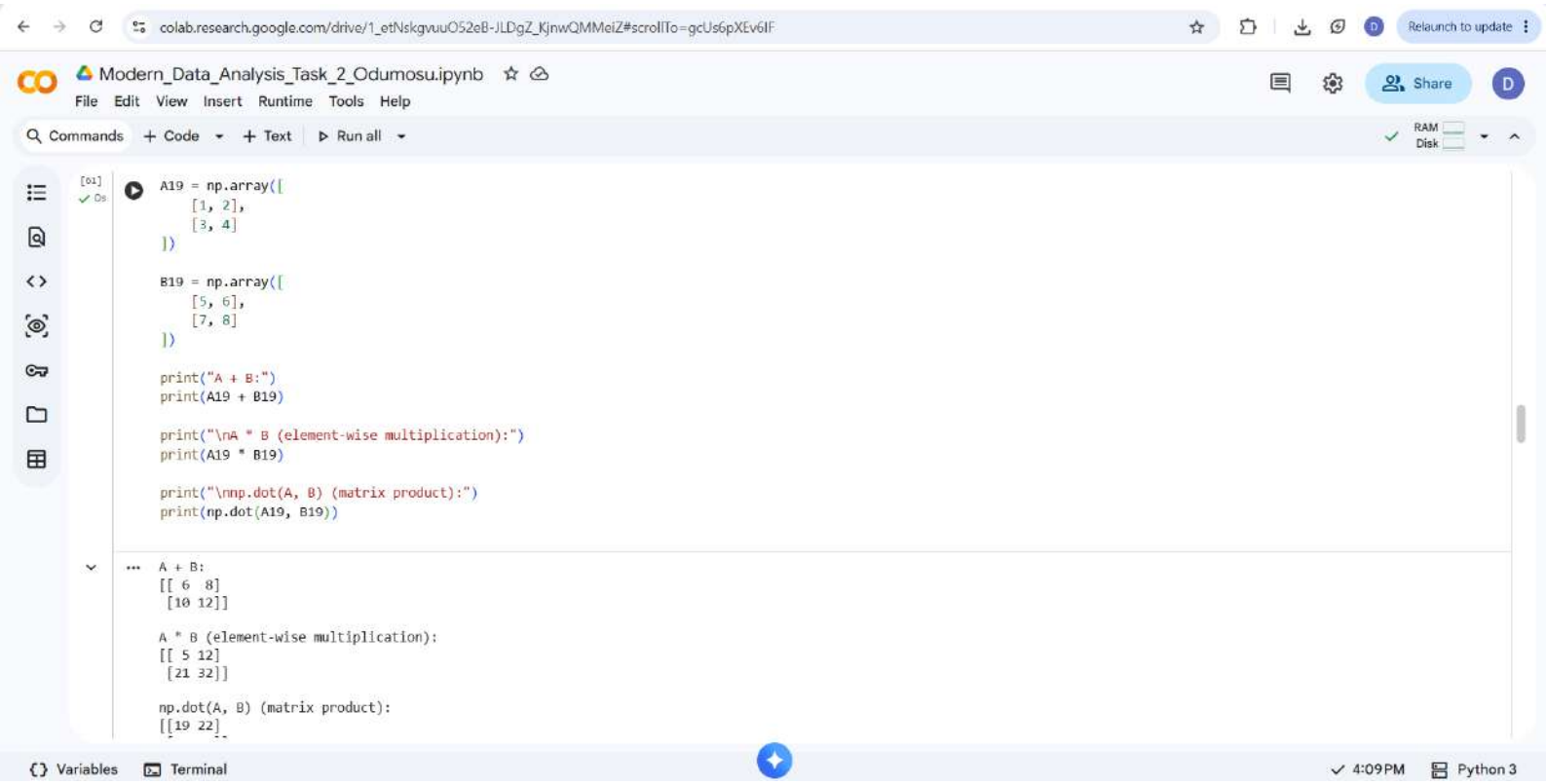
### 19. Arithmetic and matrix product (`np.dot`)

Method: To calculate the product of two compatible matrices $A$ and $B$, I multiplied each element in a row of matrix $A$ and each column of matrix $B$, and then added the resulting products to obtain each element $AB$. The matrix product is:

$$(AB)_{ij} = \sum_k A_{ik} B_{kj}$$

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[01] 0s
A19 = np.array([
[1, 2],
[3, 4]
])
B19 = np.array([
[5, 6],
[7, 8]
])
print("A + B:")
print(A19 + B19)
print("\nA * B (element-wise multiplication):")
print(A19 * B19)
print("\nnp.dot(A, B) (matrix product):")
print(np.dot(A19, B19))
A + B:
[[ 6 8]
[10 12]]
A * B (element-wise multiplication):
[[ 5 12]
[21 32]]
np.dot(A, B) (matrix product):
[[19 22]
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## Task 2.34 - Aggregation

### 27. Mean, variance, and median

Methods: I calculated the mean by taking the average of all values, I measured the variance by calculating the distance between each value, the mean and the median was identified by choosing the middle value after arranging the dataset in order.

[62]

```
data27 = np.array([2, -3, 5, 7, -1, 4, 10, -6])

mean_value = np.mean(data27)
variance_value = np.var(data27)
median_value = np.median(data27)

print("Data:")
print(data27)
print("Mean:", mean_value)
print("Variance:", variance_value)
print("Median:", median_value)
```

```
Data:
[ 2 -3  5  7 -1  4 10 -6]
Mean: 2.25
Variance: 24.9375
Median: 3.0
```

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# Task 2.35 - Comparisons, Masks, Boolean Logic

## 28. Count elements in `[-5, 5]`

I combined the NumPy Boolean conditions using the `&` operator, and used `np.count_nonzero()` to count the number of elements that satisfied all the specified conditions.

[63] 0s

```
mask28 = (data27 >= -5) & (data27 <= 5)

selected_values = data27[mask28]
count28 = np.count_nonzero(mask28)

print("Boolean mask:")
print(mask28)
print("Selected values:")
print(selected_values)
print("Count:", count28)
```

```
Boolean mask:
[ True  True  True False  True  True False False]
Selected values:
[ 2 -3  5 -1  4]
Count: 5
```

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# Task 2.36 - Adding a Column

## 33. Add column of column means

Method: I used `axis=0` to calculate the mean of each column in the matrix, then appended the calculated means as a new row to the original matrix.

[64] ✓ 0s

```
A33 = A19.copy()

column_means = np.mean(A33, axis=0)
A33_with_means = np.column_stack((A33, column_means))

print("Original matrix:")
print(A33)

print("\nColumn means:")
print(column_means)

print("\nMatrix with column means added:")
print(A33_with_means)
```

```
Original matrix:
[[1 2]
 [3 4]]
```

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```
 [3 4]]

Column means:
[2. 3.]

Matrix with column means added:
[[1. 2. 2.]
 [3. 4. 3.]]
```

## Task 2.37 - Sorting

### 38. Sort all elements ( `axis=None` )

The array was sorted using np.sort(array, `axis=None`). Setting `axis=None` allows NumPy to flatten the array and sort all elements together in ascending order.

[65]
✓ 0s

```
sorted_elements = np.sort(A33_with_means, axis=None)

print("Matrix:")
print(A33_with_means)

print("\nAll elements sorted:")
print(sorted_elements)
```

```
Matrix:
[[1. 2. 2.]
```

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```
 [3. 4. 3.]]

All elements sorted:
[1. 2. 2. 3. 3. 4.]
```

## Task 2.38 - Saving and Loading

42. Text: `np.savetxt()` / `np.loadtxt()`.

Method: I saved the data into a file and loaded it again to confirm whether the data is stored and retrieved successfully.

[66] ✓ 0s

```
file_name = "task2_array.txt"

np.savetxt(file_name, A33_with_means)

loaded_array = np.loadtxt(file_name)

print("Array saved to:", file_name)
print("\nArray loaded from the text file:")
print(loaded_array)
```

```
Array saved to: task2_array.txt

Array loaded from the text file:
[[1. 2. 2.]
 [3. 4. 3.]]
```

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## Implementation Summary

I completed the above ten assigned tasks using NumPy methods as required. The following table shows the main NumPy operations used in each of the tasks were inplemented successfully.

| Task | NumPy Operation |
|---|---|
| 4 | `np.zeros` |
| 8 | `np.random.normal` |
| 11 | Vectorized arithmetic |
| 17 | `np.linalg.norm(..., ord=np.inf)` |
| 19 | Arithmetic and `np.dot` |
| 27 | `np.mean`, `np.var`, `np.median` |
| 28 | Boolean mask and `np.count_nonzero` |
| 33 | `np.mean(axis=0)` and `np.column_stack` |
| 38 | `np.sort(axis=None)` |
| 42 | `np.savetxt` and `np.loadtxt` |

## Conclusion

This task enhanced my practical knowledge about the use of NumPy libraries for numerical data analysis.

The ten assigned tasks involved important concepts like array creation, random data generation, vectorized arithmetic, vector norms, matrix multiplication, description of data features, Boolean masking, calculation of column-based features, sorting and saving/loading

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## Conclusion

This task enhanced my practical knowledge about the use of NumPy libraries for numerical data analysis.

The ten assigned tasks involved important concepts like array creation, random data generation, vectorized arithmetic, vector norms, matrix multiplication, description of data features, Boolean masking, calculation of column-based features, sorting and saving/loading numerical data.

I learned that I can perform NumPy operations directly on entire arrays without any Python loops.

In addition, this task has shown me the difference between element-wise multiplication and matrix multiplication.

To ensure the reliability of the implementation, a constant random seed and computation validations have been used. They helped to make the results consistent and confirmed that the main output values met the necessary requirements.

In all, this task enhanced my skills in solving numerical problems using clear, effective, and meaningful NumPy methods. The obtained knowledge forms a solid basis for further research in data analysis, machine learning, computational modelling, and quantitative finance.

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